SPECTRAL INVARIANCE IS NOT BEHAVIOURAL INVARIANCE: EXACT LIMITS OF SPECTRUM-PRESERVING AND FIRST-ORDER CONSTRAINED UPDATES

SUMEET KULKARNI

Abstract. Two mechanisms are commonly proposed for preserving a property of a model while it continues to learn: restricting learnable parameters to the strictly block-upper-triangular part (the nilradical) of a fixed real Schur form, which leaves the spectrum invariant, and vetoing, at each step, the component of the update that increases a protected functional to first order. We determine exactly what each mechanism guarantees. For spectrum-preserving updates we prove that the asymptotic growth rate of every time-varying product of bounded edits equals the frozen spectral radius, so that the long-run modes are genuinely invariant; but that the set of linear readouts reachable by such edits is an affine subspace of codimension equal to the size of a single leading block (hence at least $n-2$ dimensional for generic readouts), that any family of at most that many linear probe constraints can be overwritten with an explicit norm bound, and that for every horizon k the k-step map is unbounded over the class unless an explicit polynomial condition on pairs of eigenvalues holds. For first-order vetoes we show that the commonly used Gram–Schmidt rank-k rule coincides with the correct cone projection if and only if the stored gradients are pairwise orthogonal, characterise exactly when it fails to enforce the original constraints, and prove a sharp bound: under an exact per-step veto, the increase of the protected functional is at most $\frac{\beta}{2}\sum_t \|\delta_t\|^2$, with equality attained. The veto is therefore exact in the continuum limit and its residual is a discretisation effect of order step size times path length.

1. Introduction

A recurring design goal in machine learning is to let a model keep learning while a designated property of it stays fixed. Two concrete mechanisms are the subject of this paper.

Spectral governance. Write a linear (or linearised) recurrence $x_{t+1} = Ax_t + Wu_t$ with $A = Q(D+N)Q^\top$, where Q is orthogonal, D is the quasi-diagonal part of a real Schur form, and the learnable part N is confined to the block-strictly-upper-triangular matrices $\mathfrak{n}_D$. Then $\sigma(A) = \sigma(D)$ for every value of the learnable parameters; the same holds if $N = N(u_t)$ depends on the input. Parameterisations of this kind were introduced for recurrent networks in [7], where the non-normal part is used to obtain transient dynamics, and are related to unitary and orthogonal recurrent architectures [1]. The invariance has been read as freezing the "identity" of the operator while allowing computation to change.

First-order governance. Given a protected differentiable functional f of the parameters and a proposed update δ, replace δ by its projection onto the half-space $\{v : \langle \nabla f, v\rangle \le 0\}$, or onto the cone defined by several stored gradients. This is the mechanism of GEM and A-GEM [9, 2]; related subspace methods include [3, 13, 16].

Both mechanisms are exact in what they constrain. The question we answer is what that exactness implies for behaviour. Our results are as follows.

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(1) **What spectral invariance does guarantee** (Proposition 3.1). For any sequence of bounded, possibly input-dependent edits $N_t \in \mathfrak{n}_D$, the products $\Phi_k = (D+N_k)\cdots(D+N_1)$ satisfy $\lim_k \|\Phi_k\|^{1/k} = \rho(D)$.
(2) **What it does not guarantee: readouts** (Theorem 3.3, Corollary 3.4). For a readout r, the set $\{A^\top r\}$ over the class is an explicit affine subspace whose codimension equals the total size of the leading diagonal blocks up to the first block in which $Q^\top r$ is non-zero; for generic r it has dimension at least $n-2$. Consequently any p linear probe values $r^\top A x^{(k)}$ in general position, $p \le n-2$, can be set to arbitrary targets, with $\|N\|_F$ bounded explicitly.
(3) **What it does not guarantee: finite horizons** (Theorem 3.6, Corollaries 3.7, 3.9). For a rank-one edit E supported on one off-diagonal block, $(T+cE)^k = T^k + c\,S_k(E)$ exactly, and $S_k(E)$ can be chosen non-zero if and only if an explicit polynomial $p_k(\lambda,\mu)$ is non-zero for some eigenvalue pair. The k-step map and the Kreiss constant are therefore unbounded over the class.
(4) **Rank-k vetoes** (Proposition 4.2). The Gram–Schmidt coordinate-wise rule equals the projection onto the cone of the original constraints if and only if the stored gradients are pairwise orthogonal; it enforces the original constraints if and only if all Gram–Schmidt coefficients are non-negative, and otherwise admits updates that increase some protected gradient direction to first order.
(5) **Residual under an exact veto** (Theorem 4.5, Corollaries 4.6, 4.7). Under an exact per-step veto, $f(x_T) - f(x_0) \le \frac{\beta}{2}\sum_t \|\delta_t\|^2$, with equality attained for every choice of step lengths. Hence the residual is $O(\Lambda h)$ for path length Λ and maximal step h, whereas without the veto it is $O(\Lambda)$.

Items 2 and 3 show that spectral invariance constrains only asymptotic rates; neither the one-step behaviour seen by a linear readout nor any finite-horizon response is constrained. Item 5 shows that a first-order veto is a consistent discretisation of a monotone constrained flow, so that any residual degradation under such a veto is controlled by step size and curvature rather than by the veto itself. Items 1 and 4 are, respectively, a direct consequence of the block-triangular structure (cf. the block-triangular reduction for the joint spectral radius [6]) and an elementary but, to our knowledge, unrecorded correction to a rule in practical use.

Notation. $\|\cdot\|$ is the Euclidean norm on vectors and the spectral norm on matrices; $\|\cdot\|_F$ is the Frobenius norm; $\rho(\cdot)$ is the spectral radius; $\sigma(\cdot)$ the spectrum; e_j the standard basis vectors. For a finite set of vectors, $\mathrm{cone}\{\cdot\}$ is the set of their non-negative combinations, and for a cone K, $K^\circ = \{w : \langle w, v\rangle \le 0\ \forall v \in K\}$ is its polar.

2. Preliminaries

Every $A \in \mathbb{R}^{n\times n}$ admits a real Schur decomposition $A = QTQ^\top$ with Q orthogonal and T upper quasi-triangular: block upper triangular with diagonal blocks of size 1 (real eigenvalues) or 2 (a pair of non-real complex conjugate eigenvalues) [5, Thm. 2.3.4], [4, §7.4].

Definition 2.1. Let $D = \mathrm{diag}(D_1,\dots,D_m)$ be block diagonal with blocks $D_\ell \in \mathbb{R}^{n_\ell\times n_\ell}$, $n_\ell \in \{1,2\}$, each 2×2 block having non-real eigenvalues. Let $I_1,\dots,I_m$ be the corresponding consecutive index sets and $\mathrm{blk}(i) = \ell$ for $i \in I_\ell$. The *nilradical* of D is

$$\mathfrak{n}_D = \{N \in \mathbb{R}^{n\times n} :\ N_{ij} = 0 \text{ unless } \mathrm{blk}(i) < \mathrm{blk}(j)\}.$$

For an orthogonal Q, the *spectrum-preserving class* is $\mathcal{A}_{D,Q} = \{Q(D+N)Q^\top :\ N \in \mathfrak{n}_D\}$.

For a matrix M partitioned conformally with $I_1,\dots,I_m$, we write $M_{(p,q)}$ for its (p,q) block. Every $T \in D + \mathfrak{n}_D$ is block upper triangular with $T_{(\ell,\ell)} = D_\ell$.

The following facts are standard; we record them with proofs for completeness.

Proposition 2.2 (Standard)**.** *Let $N, N_1, \dots, N_k \in \mathfrak{n}_D$.*

(1) $\sigma(D+N) = \sigma(D)$.
(2) $I + \mathfrak{n}_D$ *is a group under multiplication, and for* $U \in I + \mathfrak{n}_D$ *and* $T \in D + \mathfrak{n}_D$, $\sigma(UT) = \sigma(D)$.
(3) *The product* $\Phi_k = (D+N_k)\cdots(D+N_1)$ *is block upper triangular with* $(\Phi_k)_{(\ell,\ell)} = D_\ell^k$*; in particular* $\sigma(\Phi_k) = \{\lambda^k : \lambda \in \sigma(D)\}$.

Proof. Block upper triangular matrices with respect to a fixed partition form an algebra, and the diagonal block of a product is the product of the diagonal blocks. The spectrum of a block upper triangular matrix is the union of the spectra of its diagonal blocks. (1) and (3) follow. For (2), $(I+N)(I+N') = I+N+N'+NN' \in I+\mathfrak{n}_D$, and $(I+N)^{-1} = \sum_{j=0}^{m-1}(-N)^j \in I+\mathfrak{n}_D$ because $N^m = 0$; finally UT has diagonal blocks $I \cdot D_\ell$. □

Because $\|QMQ^\top\| = \|M\|$ and $r^\top QMQ^\top x = (Q^\top r)^\top M(Q^\top x)$, all statements below may be read either in Schur coordinates or in the original coordinates.

3. Spectral governance

3.1. What is preserved: asymptotic rates.

Proposition 3.1. *Let $N_t \in \mathfrak{n}_D$ with $\sup_t \|N_t\| \le \nu < \infty$ and $\Phi_k = (D+N_k)\cdots(D+N_1)$. Then $\lim_{k\to\infty} \|\Phi_k\|^{1/k} = \rho(D)$.*

Proof. Lower bound. By Proposition 2.2(3), for every ℓ, $\|\Phi_k\| \ge \|(\Phi_k)_{(\ell,\ell)}\| = \|D_\ell^k\| \ge \rho(D_\ell)^k$. Choosing ℓ with $\rho(D_\ell) = \rho(D)$ gives $\liminf_k \|\Phi_k\|^{1/k} \ge \rho(D)$.

Upper bound. Fix $\varepsilon > 0$. For each ℓ choose a vector norm $\|\cdot\|_\ell$ on $\mathbb{R}^{n_\ell}$ whose induced norm satisfies $\|D_\ell\|_\ell \le \rho(D_\ell) + \varepsilon$ [5, §5.6]. On $\mathbb{R}^n$ define $\|x\|_* = \max_\ell \|x_{I_\ell}\|_\ell$. For a block matrix B the induced norm satisfies $\|B\|_* \le \max_p \sum_q \|B_{(p,q)}\|_{q\to p}$, where $\|\cdot\|_{q\to p}$ is the operator norm from $(\mathbb{R}^{n_q}, \|\cdot\|_q)$ to $(\mathbb{R}^{n_p}, \|\cdot\|_p)$. All norms on finite-dimensional spaces are equivalent, so there is $C < \infty$, depending only on D, ν and ε, with $\|(D+N_t)_{(p,q)}\|_{q\to p} \le C$ for all t, p, q. For $s \in (0,1]$ let $S_s = \operatorname{diag}(s^1 I_{n_1}, \dots, s^m I_{n_m})$. Then

$$\big(S_s^{-1}(D+N_t)S_s\big)_{(p,q)} = s^{\,q-p}(D+N_t)_{(p,q)},$$

which vanishes for $q < p$, equals D_p for $q = p$, and has norm at most sC for $q > p$. Hence $\|S_s^{-1}(D+N_t)S_s\|_* \le \rho(D) + \varepsilon + (m-1)sC$. Choose s with $(m-1)sC \le \varepsilon$. Since $\Phi_k = S_s\big(\prod_t S_s^{-1}(D+N_t)S_s\big)S_s^{-1}$ and $\|\cdot\|_*$ is equivalent to $\|\cdot\|$, there is $\kappa < \infty$ independent of k with $\|\Phi_k\| \le \kappa(\rho(D) + 2\varepsilon)^k$. Thus $\limsup_k \|\Phi_k\|^{1/k} \le \rho(D) + 2\varepsilon$ for every $\varepsilon > 0$. □

Remark 3.2. Proposition 3.1 is the precise sense in which spectrum-preserving edits fix the "long-run modes": for bounded, even input-dependent, edits the exponential growth or decay rate of the state transition is exactly $\rho(D)$. The bound ν is essential to the upper bound only through the constant κ; Theorem 3.6 shows that κ cannot be bounded uniformly over the class.

3.2. What is not preserved: linear readouts.

For $r \in \mathbb{R}^n \setminus \{0\}$ put $a = Q^\top r$,

$$\beta(r) = \min\{\mathrm{blk}(i) : a_i \neq 0\}, \qquad J(r) = \{j : \mathrm{blk}(j) > \beta(r)\}, \qquad \mathbb{R}^J = \operatorname{span}\{e_j : j \in J(r)\}.$$

Theorem 3.3. *For every $r \neq 0$,*

$$\{A^\top r : A \in \mathcal{A}_{D,Q}\} = Q\big(D^\top a + \mathbb{R}^{J(r)}\big),$$

an affine subspace of dimension $|J(r)| = n - \sum_{\ell \le \beta(r)} n_\ell$. In particular, if $a_{I_1} \neq 0$, which holds for all r outside a proper linear subspace, the dimension is $n - n_1 \ge n - 2$.

Proof. For $A = Q(D+N)Q^\top$ we have $A^\top r = Q(D^\top a + N^\top a)$, so it suffices to show $\{N^\top a : N \in \mathfrak{n}_D\} = \mathbb{R}^{J(r)}$. Let $R_j = \{i : \mathrm{blk}(i) < \mathrm{blk}(j)\}$. Then $(N^\top a)_j = \sum_{i \in R_j} N_{ij} a_i$, and distinct j involve disjoint sets of free entries of N. If $j \notin J(r)$ then $\mathrm{blk}(i) < \mathrm{blk}(j) \le \beta(r)$ for all $i \in R_j$, so $a_i = 0$ there and $(N^\top a)_j = 0$. If $j \in J(r)$ then $R_j \supseteq I_{\beta(r)}$, so $a_{R_j} \neq 0$ and $(N^\top a)_j$ ranges over all of $\mathbb{R}$. Hence the image is exactly $\mathbb{R}^{J(r)}$. The dimension count is immediate, and $a_{I_1} \neq 0$ fails only on the subspace $\{r : (Q^\top r)_{I_1} = 0\}$ of codimension n_1. □

Corollary 3.4. *Let $x^{(1)}, \dots, x^{(p)} \in \mathbb{R}^n$, $y^{(k)} = Q^\top x^{(k)}$, and let $Y_J \in \mathbb{R}^{p \times |J(r)|}$ have rows $(y^{(k)})_{J(r)}$. If $\operatorname{rank} Y_J = p$, then for every $c \in \mathbb{R}^p$ there is $N \in \mathfrak{n}_D$ with*

$$r^\top Q(D+N)Q^\top x^{(k)} = c_k \quad (k = 1, \dots, p), \qquad \|N\|_F \le \frac{\|c - b\|}{\sigma_p(Y_J)\, \|a_{I_{\beta(r)}}\|},$$

where $b_k = r^\top Q D Q^\top x^{(k)}$ and σ_p denotes the p-th singular value. The rank condition holds for probes in general position whenever $p \le |J(r)|$.

Proof. We have $r^\top Q(D+N)Q^\top x^{(k)} = b_k + (N^\top a)^\top y^{(k)}$. By Theorem 3.3, $N^\top a$ may be any w supported on $J(r)$, so the constraints read $Y_J w_J = c - b$, solvable with $\|w\| \le \|c-b\|/\sigma_p(Y_J)$ by the minimum-norm solution. Given such w, set $N_{ij} = a_i w_j / \|a_{R_j}\|^2$ for $j \in J(r)$, $i \in R_j$, and $N_{ij} = 0$ otherwise. Then $N \in \mathfrak{n}_D$, $(N^\top a)_j = w_j$, and $\|N\|_F^2 = \sum_{j \in J(r)} w_j^2 / \|a_{R_j}\|^2 \le \|w\|^2 / \|a_{I_{\beta(r)}}\|^2$ since $R_j \supseteq I_{\beta(r)}$. □

Remark 3.5. In particular, a linear functional that is positive on a set of probe inputs (for instance a logit difference encoding a protected behaviour) can be made negative on all of them, by an edit in $\mathfrak{n}_D$, while $\sigma(A)$ is unchanged. Proposition 2.2 certifies this edit as spectrum-preserving.

3.3. **What is not preserved: finite horizons.** For $k \ge 1$ define the polynomial $p_k(\lambda, \mu) = \sum_{i=0}^{k-1} \lambda^i \mu^{k-1-i}$, so that $p_k(\lambda, \mu) = (\lambda^k - \mu^k)/(\lambda - \mu)$ for $\lambda \neq \mu$ and $p_k(\lambda, \lambda) = k\lambda^{k-1}$.

Theorem 3.6. *Let $T \in D + \mathfrak{n}_D$, let $1 \le p < q \le m$, and let $E \in \mathfrak{n}_D$ be supported in the block (p, q), with $E_{(p,q)} = Z$. Then for all $c \in \mathbb{R}$ and $k \ge 1$,*

$$(T + cE)^k = T^k + c\, S_k(E), \qquad S_k(E) = \sum_{i=0}^{k-1} T^i E T^{k-1-i}, \tag{1}$$

and $S_k(E)_{(p,q)} = L_k(Z) := \sum_{i=0}^{k-1} D_p^i Z D_q^{k-1-i}$. Moreover, $L_k \not\equiv 0$ if and only if there exist $\lambda \in \sigma(D_p)$ and $\mu \in \sigma(D_q)$ with $p_k(\lambda, \mu) \neq 0$. In that case, choosing Z with $L_k(Z) \neq 0$,

$$\|(T + cE)^k\| \;\ge\; |c|\, \|L_k(Z)\| - \|T^k\| \;\xrightarrow[|c| \to \infty]{}\; \infty,$$

while $\sigma(T + cE) = \sigma(D)$ for all c.

Proof. Expand $(T + cE)^k$ as a sum over words in T and cE. Every power T^j is block upper triangular, so $(T^j)_{(q,p)} = 0$ because $q > p$. Since E is supported in block (p, q), ET^jE has the single block $Z(T^j)_{(q,p)}Z = 0$; hence every word containing E at least twice vanishes, which gives (1). For the (p, q) block of $T^i E T^{k-1-i}$, only the block (p, q) of E contributes, giving $(T^i)_{(p,p)} Z (T^{k-1-i})_{(q,q)} = D_p^i Z D_q^{k-1-i}$ by Proposition 2.2(3).

Each D_ℓ is diagonalisable over $\mathbb{C}$: trivially if $n_\ell = 1$, and because its two eigenvalues are distinct (a non-real conjugate pair) if $n_\ell = 2$. Write $D_p = V \Lambda V^{-1}$ and $D_q = W M W^{-1}$ with Λ, M diagonal. For complex Z, put $X = V^{-1} Z W$; then $V^{-1} L_k(Z) W$ has entries $X_{ab}\, p_k(\Lambda_{aa}, M_{bb})$. Hence the complexification of L_k vanishes identically if and only if $p_k(\lambda, \mu) = 0$ for all $\lambda \in \sigma(D_p)$,

$\mu \in \sigma(D_q)$. Since L_k is a real-linear map commuting with entrywise conjugation, it vanishes on real matrices if and only if its complexification vanishes. The final inequality follows from (1) and $\|S_k(E)\| \geq \|S_k(E)_{(p,q)}\|$; the spectral statement is Proposition 2.2(1). □

Corollary 3.7. *Suppose D has eigenvalues $\lambda \in \sigma(D_p)$, $\mu \in \sigma(D_q)$ with $p < q$ such that either $\lambda = \mu \neq 0$, or $\mu = 0 \neq \lambda$, or $\lambda \neq \mu$, $\mu \neq 0$ and λ/μ is not a root of unity. Then for every horizon $k \geq 1$, $\sup\{\|A^k\| : A \in \mathcal{A}_{D,Q}\} = \infty$.*

Proof. In each case $p_k(\lambda, \mu) \neq 0$ for all $k \geq 1$: $p_k(\lambda, \lambda) = k\lambda^{k-1}$; $p_k(\lambda, 0) = \lambda^{k-1}$; and for $\lambda \neq \mu$, $p_k(\lambda, \mu) = 0$ iff $(\lambda/\mu)^k = 1$. Apply Theorem 3.6 with $T = D$ and note $\|QA^kQ^\top\| = \|A^k\|$. □

Remark 3.8. The exceptions in Theorem 3.6 are genuine. If $D_p = \lambda$ and $D_q = -\lambda$ are scalars, then $L_2 \equiv 0$, and indeed $(D + cE)^2 = D^2$ for all c by (1).

Recall the Kreiss constant $\mathcal{K}(A) = \sup_{|z|>1}(|z|-1)\|(zI-A)^{-1}\|$ and the Kreiss matrix theorem with Spijker's constant [8, 14, 15]:

$$\mathcal{K}(A) \;\leq\; \sup_{k\geq 0} \|A^k\| \;\leq\; e\,n\,\mathcal{K}(A). \tag{2}$$

Corollary 3.9. *Under the hypotheses of Corollary 3.7 and $\rho(D) < 1$, every $A \in \mathcal{A}_{D,Q}$ satisfies $\|A^k\| \to 0$ as $k \to \infty$, yet $\sup\{\mathcal{K}(A) : A \in \mathcal{A}_{D,Q}\} = \infty$.*

Proof. The first claim follows from $\rho(A) = \rho(D) < 1$. For the second, by (2), $\mathcal{K}(A) \geq \|A^k\|/(en)$ for any fixed k, and the right-hand side is unbounded over the class by Corollary 3.7. □

Remark 3.10. Theorems 3.3 and 3.6 together with Proposition 3.1 give a complete qualitative picture. Spectrum-preserving edits fix the asymptotic rate of every state transition. They leave free a subspace of linear readouts of dimension at least $n - 2$, and an amount of finite-horizon amplification that is linear and unbounded in the edit size. Any behaviour that is read out at a finite horizon, or through a linear functional, is therefore not protected by spectral invariance alone.

4. First-order governance

Let $f : \mathbb{R}^d \to \mathbb{R}$ be a protected functional of the parameters, differentiable, and let $\delta \in \mathbb{R}^d$ be a proposed update produced by any optimiser.

Lemma 4.1 (Standard)**.** *For $g \neq 0$ and $u = g/\|g\|$, the map $\delta \mapsto \delta - \max(\langle \delta, u\rangle, 0)\, u$ is the Euclidean projection onto $H_g = \{v : \langle g, v\rangle \leq 0\}$.*

Proof. If $\langle \delta, u\rangle \leq 0$ then $\delta \in H_g$. Otherwise the projection onto a closed half-space removes the normal component of the excess, which is $\langle \delta, u\rangle u$. □

For stored gradients $g_1, \ldots, g_k$ (with g_1 the most recent), the constraint set is the polyhedral cone $K_G = \{v : \langle g_i, v\rangle \leq 0,\ i = 1, \ldots, k\}$. Its projection is $P_{K_G}(\delta) = \delta - G^\top\lambda^\star$ with $\lambda^\star \in \arg\min_{\lambda\geq 0} \frac{1}{2}\|\delta - G^\top\lambda\|^2$, by Moreau's decomposition $\delta = P_K(\delta) + P_{K^\circ}(\delta)$ and $K_G^\circ = \mathrm{cone}\{g_i\}$ [10, 12]. A cheaper rule orthonormalises $g_1, \ldots, g_k$ by Gram–Schmidt into $u_1, \ldots, u_k$ and applies Lemma 4.1 coordinate-wise:

$$V_{\mathrm{GS}}(\delta) = \delta - \sum_{i=1}^{k} \max(\langle \delta, u_i\rangle, 0)\, u_i. \tag{3}$$

Write $K_U = \{v : \langle u_i, v\rangle \leq 0,\ i = 1, \ldots, k\}$ and $\gamma_{ij} = \langle g_j, u_i\rangle$ for $i < j$ (the Gram–Schmidt coefficients).

Proposition 4.2. *Let $g_1, \ldots, g_k$ be linearly independent. Then:*

(1) $V_{\mathrm{GS}} = P_{K_U}$; *in particular, applying the k corrections sequentially or simultaneously gives the same result.*
(2) $K_U \subseteq K_G$ *if and only if* $\gamma_{ij} \geq 0$ *for all* $i < j$.
(3) $K_U = K_G$ *if and only if* $g_1, \ldots, g_k$ *are pairwise orthogonal.*
(4) *If* $\gamma_{ij} < 0$ *for some* $i < j$, *there is* δ *with* $V_{\mathrm{GS}}(\delta) = \delta$ *and* $\langle g_{j'}, \delta \rangle > 0$ *for some* j'; *that is, the rule passes an update that increases* f *to first order along a stored gradient.*

Proof. (1) Write $\delta = \sum_i \langle \delta, u_i \rangle u_i + \delta_\perp$. Minimising $\|v - \delta\|^2$ over $v \in K_U$ decouples across the orthonormal coordinates, giving $\langle v, u_i \rangle = \min(\langle \delta, u_i \rangle, 0)$ and $v_\perp = \delta_\perp$, which is (3). Sequential application gives the same result because subtracting a multiple of u_i leaves $\langle \cdot, u_{i'} \rangle$ unchanged for $i' \neq i$.

(2) Both cones are closed and convex, so $K_U \subseteq K_G$ iff $K_G^\circ \subseteq K_U^\circ$ [12, Thm. 14.1], i.e. iff each $g_j \in \mathrm{cone}\{u_1, \ldots, u_k\}$. By construction $g_j = \sum_{i<j} \gamma_{ij} u_i + \|r_j\| u_j$, where r_j is the Gram–Schmidt residual, which is non-zero by independence. This expansion is unique because the u_i are orthonormal, so $g_j \in \mathrm{cone}\{u_i\}$ iff $\gamma_{ij} \geq 0$ for all $i < j$.

(3) If the g_i are pairwise orthogonal then $u_i = g_i / \|g_i\|$ and the cones coincide. Conversely, $K_U = K_G$ implies $\mathrm{cone}\{u_i\} = \mathrm{cone}\{g_i\}$. A cone generated by linearly independent vectors is simplicial, and its extreme rays are exactly the rays of its generators. Hence each u_i is a positive multiple of some g_j. If $j < i$ this contradicts $u_i \perp g_j$. If $j > i$ it contradicts $g_j \notin \mathrm{span}\{g_1, \ldots, g_i\} \ni u_i$. So $u_i \propto g_i$, which forces $\langle g_i, u_l \rangle = 0$, i.e. $g_i \perp g_l$, for all $l < i$.

(4) By (2), $K_U \not\subseteq K_G$, so some $\delta \in K_U \setminus K_G$ exists. Then $V_{\mathrm{GS}}(\delta) = P_{K_U}(\delta) = \delta$ and $\langle g_{j'}, \delta \rangle > 0$ for some j'. □

Example 4.3. In $\mathbb{R}^2$ take $g_1 = e_1$ and $g_2 = (-e_1 + e_2)/\sqrt{2}$, so $\gamma_{12} = -1/\sqrt{2} < 0$, $u_1 = e_1$ and $u_2 = e_2$. For $\delta = (-1, 0.9)$, $V_{\mathrm{GS}}(\delta) = (-1, 0)$ and $\langle g_2, (-1, 0) \rangle = 1/\sqrt{2} > 0$.

Remark 4.4. When $\gamma_{ij} \geq 0$ for all $i < j$ but the g_i are not orthogonal, $K_U \subsetneq K_G$ and V_{GS} is valid but conservative: $\|\delta - V_{\mathrm{GS}}(\delta)\| \geq \|\delta - P_{K_G}(\delta)\|$ for every δ, with strict inequality for some δ. In either case the exact rule is P_{K_G}, a k-variable non-negative least-squares problem.

We now quantify what a veto achieves along a trajectory. Say that a trajectory $x_{t+1} = x_t + \delta_t$, $t = 0, \ldots, T-1$, satisfies an *exact per-step veto* if $\langle \nabla f(x_t), \delta_t \rangle \leq 0$ for every t. This holds, for instance, if each δ_t is the output of Lemma 4.1 or of P_{K_G} with $g_1 = \nabla f(x_t)$.

Theorem 4.5. *Let ∇f be β-Lipschitz on a convex set containing all segments $[x_t, x_{t+1}]$. If the trajectory satisfies an exact per-step veto, then*

$$f(x_T) - f(x_0) \;\leq\; \frac{\beta}{2} \sum_{t=0}^{T-1} \|\delta_t\|^2. \tag{4}$$

The bound is sharp: for $d \geq 2$, $f(x) = \frac{\beta}{2}\|x\|^2$, $x_0 = 0$ and any step lengths $s_0, \ldots, s_{T-1} \geq 0$, there is a trajectory with $\|\delta_t\| = s_t$ satisfying an exact per-step veto with equality in (4).

Proof. By the descent lemma [11, Lemma 1.2.3], $f(x_t + \delta_t) \leq f(x_t) + \langle \nabla f(x_t), \delta_t \rangle + \frac{\beta}{2}\|\delta_t\|^2 \leq f(x_t) + \frac{\beta}{2}\|\delta_t\|^2$. Summing over t gives (4).

For sharpness, take δ_0 arbitrary with $\|\delta_0\| = s_0$; the veto condition holds since $\nabla f(0) = 0$. For $t \geq 1$ choose $\delta_t \perp x_t$ with $\|\delta_t\| = s_t$, which is possible since $d \geq 2$. Then $\langle \nabla f(x_t), \delta_t \rangle = \beta \langle x_t, \delta_t \rangle = 0$, and by Pythagoras $\|x_T\|^2 = \sum_t s_t^2$, so $f(x_T) - f(x_0) = \frac{\beta}{2} \sum_t s_t^2$. □

Corollary 4.6 (Continuum limit)**.** *Under the hypotheses of Theorem 4.5, let $\Lambda = \sum_t \|\delta_t\|$ and $h = \max_t \|\delta_t\|$. Then $f(x_T) - f(x_0) \leq \frac{\beta}{2}\Lambda h$. Hence for any sequence of vetoed trajectories with path length at most Λ and $h \to 0$, $\limsup (f(x_T) - f(x_0)) \leq 0$. Without the veto, the*

corresponding bound is $f(x_T) - f(x_0) \le G\Lambda + \frac{\beta}{2}\Lambda h$ *with* $G = \sup_t \|\nabla f(x_t)\|$*, and the first term does not vanish as* $h \to 0$*.*

Proof. $\sum_t \|\delta_t\|^2 \le h \sum_t \|\delta_t\| = h\Lambda$. The unvetoed bound follows from the descent lemma and $\langle \nabla f(x_t), \delta_t \rangle \le G\|\delta_t\|$. □

Corollary 4.7 (Stale references)**.** *Suppose instead* $\langle \nabla f(z_t), \delta_t \rangle \le 0$ *for reference points* z_t*, for instance the point at which the stored gradient was computed. Then*

$$f(x_T) - f(x_0) \;\le\; \beta \sum_t \|x_t - z_t\|\,\|\delta_t\| + \frac{\beta}{2}\sum_t \|\delta_t\|^2.$$

Proof. $\langle \nabla f(x_t), \delta_t \rangle \le \langle \nabla f(x_t) - \nabla f(z_t), \delta_t \rangle \le \beta \|x_t - z_t\| \|\delta_t\|$; combine with the descent lemma as above. □

Remark 4.8. Theorem 4.5 and Corollary 4.6 isolate what a first-order veto controls. It removes the term that is first order in the path, $\sum_t \langle \nabla f(x_t), \delta_t \rangle$, entirely. It leaves a term of order Λh that depends only on curvature and step length, and the veto has no effect on that term. For a fixed path, halving the step size at least halves the bound in (4). Whenever per-step updates have norm bounded below, as for optimisers with per-coordinate normalisation, the residual accumulates linearly in the number of steps.

5. Discussion

The results separate two notions that are easily conflated. *Exact invariance of a quantity* (the spectrum; the sign of the first-order change of f) holds by construction for both mechanisms. *Invariance of behaviour* holds only to the extent established above:

- Spectrum-preserving edits fix asymptotic rates (Proposition 3.1). They do not fix linear readouts (Theorem 3.3, Corollary 3.4) or finite-horizon responses (Theorem 3.6).
- First-order vetoes fix the sign of the first-order change. They leave a curvature-driven residual that is sharp (Theorem 4.5) and vanishes only in the continuum limit (Corollary 4.6).

Both mechanisms protect only what they are defined on. A first-order veto acts on a chosen functional f and says nothing about functionals that are not constrained. A spectral constraint acts on a chosen linear operator and says nothing about readouts of it. Guarantees about behaviour therefore require constraining the behaviour itself, for instance an output functional on the inputs of interest, or controlling step size and curvature through Theorem 4.5.

References

[1] M. Arjovsky, A. Shah, and Y. Bengio, *Unitary evolution recurrent neural networks*, Proceedings of the 33rd International Conference on Machine Learning (ICML), 2016.

[2] A. Chaudhry, M. Ranzato, M. Rohrbach, and M. Elhoseiny, *Efficient lifelong learning with A-GEM*, International Conference on Learning Representations (ICLR), 2019.

[3] M. Farajtabar, N. Azizan, A. Mott, and A. Li, *Orthogonal gradient descent for continual learning*, Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS), 2020.

[4] G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th ed., Johns Hopkins University Press, Baltimore, 2013.

[5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, Cambridge, 2013.

[6] R. M. Jungers, *The Joint Spectral Radius: Theory and Applications*, Lecture Notes in Control and Information Sciences, vol. 385, Springer, Berlin, 2009.

[7] G. Kerg, K. Goyette, M. P. Touzel, G. Gidel, E. Vorontsov, Y. Bengio, and G. Lajoie, *Non-normal recurrent neural network (nnRNN): learning long time dependencies while improving expressivity with transient dynamics*, Advances in Neural Information Processing Systems 32 (NeurIPS), 2019.

[8] H.-O. Kreiss, *Über die Stabilitätsdefinition für Differenzengleichungen die partielle Differentialgleichungen approximieren*, BIT Numerical Mathematics **2** (1962), 153–181.
[9] D. Lopez-Paz and M. Ranzato, *Gradient episodic memory for continual learning*, Advances in Neural Information Processing Systems 30 (NeurIPS), 2017.
[10] J.-J. Moreau, *Décomposition orthogonale d'un espace hilbertien selon deux cônes mutuellement polaires*, Comptes Rendus de l'Académie des Sciences de Paris **255** (1962), 238–240.
[11] Y. Nesterov, *Introductory Lectures on Convex Optimization: A Basic Course*, Kluwer Academic Publishers, Boston, 2004.
[12] R. T. Rockafellar, *Convex Analysis*, Princeton University Press, Princeton, 1970.
[13] G. Saha, I. Garg, and K. Roy, *Gradient projection memory for continual learning*, International Conference on Learning Representations (ICLR), 2021.
[14] M. N. Spijker, *On a conjecture by LeVeque and Trefethen related to the Kreiss matrix theorem*, BIT Numerical Mathematics **31** (1991), 551–555.
[15] L. N. Trefethen and M. Embree, *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*, Princeton University Press, Princeton, 2005.
[16] S. Wang, X. Li, J. Sun, and Z. Xu, *Training networks in null space of feature covariance for continual learning*, Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2021.

[Affiliation]
Email address: [email]